

The
ART

of

WEB DEVELOPMENT

A handbook by

SUDIPTO DAS



This is what we consider AI as, a complete *magic*

Although it is, but to a certain extent. There are still many areas where using AI tools is not so much effective and one of them is **web development**.

You can

Vibe code an application in 5 minutes

Launch a small idea

Create UI designs in figma

Automate small to medium tasks

But you can't

Build an application with multiple complexities

Deploy AI code in production

Generate entirely bug free code

Automate task with multiple steps or several functions



What's so special in

WEB DEVELOPMENT



Learning web development is like playing with building blocks. You start by building the base and then step by step, pieces by pieces you create an entire huge block structure.

Similarly in web development you learn about the web, how it works behind the scene, how an entire website works, what are static and dynamic contents of a website, realtime events, animations, interactions, UI and lot more.

AI tools like : Cursor, Github Copilot, Windsurf, Lovable are so much popular for vibe coding and most effective coding assistants and at some extent they really help well and saves time but you can't always guarantee that they will generate bug free code. At the end you have to check and debug for any issues and vulnerabilities but now surprise, If you don't really know how things work internally then you have to spend your rest of the time by telling the AI to fix it.

LEVEL

01

Frontend

Think of it as the visible part or the section which the users see and interact with. Whenever you open zomato or amazon, the entire application is a frontend that you see from the outside. The animations, the buttons, the colors, the graphics, all the UI you see is the part of frontend.

The frontend part is mostly built using technologies like : HTML, CSS, JavaScript or in today's world we have frameworks and libraries like : Angular, ReactJs, Svelte, TailwindCSS etc which make it easy for us to built these frontend UIs.



LEVEL 02

Backend

Now, the backend is like a mastermind, it controls the entire website logic and operations but you can't see it, obviously it is in back so the name is **backend**

Operations like database connectivity, payment operations, condition check, password hashing, CRUD (create, read, update, delete), mail sending and bla bla bla...all these comes under backend.

All of we have used zomato, atleast once, so have you ever thought how zomato knows who am I, or how my order correctly delivers on my address, or even how everytime it notifies my where is my delivery right now. All of this is possible because of backend communication.

Moral : Without backend, the frontend is like a wedding party but without food service, **incomplete & pointless**.



LEVEL
03

DevOps

Now that you know about frontend and backend, let's talk about the bridge that connects them smoothly — DevOps.

Think of DevOps as that super-organized event manager who makes sure everything runs perfectly.

Imagine this — you're using Zomato, and suddenly they roll out a new design or feature, and it's there instantly without server crash or any issue. That's DevOps magic right there — keeping updates rolling like butter on hot toast

DevOps isn't just about coding, it's about teamwork, automation, and making life easier for developers. It makes sure that when someone updates a small feature or fixes a bug, the whole website doesn't explode (thankfully).

It handles stuff like deployment, testing, monitoring, CI/CD (continuous integration and delivery), server management, and all those behind-the-scenes things that make apps run faster, smoother, and safer.



Moral of the STORY

AI can be your assistant, a helping hand, a tool that saves time but for making meaningful applications that actually serves any purpose, that can be easily maintained, also gets handled by other developers human intervention is still required.

AI tools can help you write test cases, create workflows, default configuration but for total control in development and in production, no one is best other than any human.

Powerful
AI is ~~Perfect~~

Frontend

HTML
CSS
JavaScript
TypeScript
TailwindCSS
ReactJs / AngularJs
Redux / Zustand



Backend

NodeJs , ExpressJs
Django / Flask
SpringBoot
C# , .NET



Deploy

Vercel
Heroku
AWS
Google Cloud
Microsoft Azure
Digital Ocean



Databases

SQL

MySQL
MS SQL
PostgreSQL
MariaDB
SQLite

NoSQL

MongoDB
Cassandra
Redis
Couchbase
NeonDB



Infra

Linux
Github Actions / Gitlab CI CD
Docker, Kubernetes
Grafana
Git, Github
Ansible, Chef



**Keep
Learning**